

IPv4

IPv4 stands for Internet Protocol Version 4. It is the fourth version of the Internet Protocol used for identifying devices on a network.

IPv4 uses 32-bit addresses and is written in dotted decimal notation.

Example:
192.168.1.1

Each IPv4 address contains four parts called octets. Every octet contains 8 bits.

IPv4 provides approximately 4.3 billion unique addresses.

Main Functions:

- Identifies devices on network
- Helps in routing data
- Enables communication between devices

Advantages:

- Simple and widely used
- Easy configuration
- Supported by almost all networks

Disadvantage:

- Limited number of addresses

Real Life Example:

Like every house has a unique address, every device on the internet has an IPv4 address.

The IPv4 Classes IPv4 addresses are divided into classes based on the size of the network:

- **Class A:** For massive networks. The first octet is between 0 and 127.
- **Class B:** For medium-sized networks. The first octet is between 128 and 191.
- **Class C:** For small networks. The first octet is between 192 and 223.

2. IPv4 vs. IPv6: The Core Difference

Real-Life Example: Imagine IPv4 as a phone number system with only 7 digits. Eventually, as more people get phones, you run out of numbers! That is exactly what happened with the 4.3 billion IPv4 addresses. IPv6 was created as the "new phone system" with way more digits (128-bit compared to IPv4's 32-bit) so we will never run out of addresses for our smartphones, smart TVs, and laptops.

IPv6

IPv6 stands for Internet Protocol Version 6. It was introduced to solve the shortage of IPv4 addresses.

IPv6 uses 128-bit addresses and provides a huge number of unique addresses.

Example:

2001:db8:85a3::8a2e:370:7334

IPv6 improves security, routing, and network efficiency.

Features:

- 128-bit addressing
- Very large address space
- Better security support
- Faster routing

Advantages:

- Unlimited address availability
- Improved performance
- Better support for modern networks

Real Life Example:

IPv6 is like expanding a small town into a huge city with millions of new addresses.

3. Subnetting (The Class Problems)

Subnetting can seem tricky, but it is just a way of slicing a big pie into smaller, manageable slices.

- **What is it?** Subnetting is the process of dividing a larger network into smaller sub-networks.
- **Why do it?** It helps organizations use their IP addresses efficiently and improves network security and speed.
- *Real-Life Example:* Imagine a massive corporate office. If all 1,000 employees print to one giant pool of printers, it's chaos. Subnetting is like building walls to create an HR department, a Sales department, and an IT department, each with its own local printers.

Difference Between IPv4 and IPv6

IPv4

32-bit address

Decimal format

Limited addresses

Older version

Less secure

IPv6

128-bit address

Hexadecimal format

Very large address space

Newer version

Better security



TCP/IP Protocol Suite

What is TCP/IP

TCP/IP is a set of rules that devices follow to communicate over the internet. Think of it as a universal language that every computer, phone, and server speaks so they can understand each other. TCP and IP are the two most important protocols in this set, and each one handles a different job.

TCP — Transmission Control Protocol

What it does TCP makes sure your data reaches the other side completely and correctly. It is like a responsible delivery service that checks every package and resends anything that gets lost.

How it connects Before sending any data, TCP does a Three-Way Handshake to set up the connection.

The sender says "Can we connect?" — this is called SYN. The receiver replies "Yes, I am ready" — this is SYN-ACK. The sender confirms "Great, let us start" — this is ACK. After this, data transfer begins.

Key points TCP breaks data into small pieces called segments and numbers them so they can be reassembled in the right order at the destination. If any piece goes missing, TCP asks for it again. This makes it reliable but slightly slower because of all the checking.

Used for Web browsing, email, file downloads — anything where accuracy matters more than speed.

IP — Internet Protocol

What it does IP gives every device a unique address and is responsible for getting data from one device to another across the network. It decides the route the data will take.

How it works Data is split into packets. Each packet carries a source address and a destination address. Routers across the network read these addresses and forward each packet closer to its destination. Different

packets of the same message can take different routes and meet back up at the end.

Key point IP does not guarantee delivery. It just does its best to forward packets. Guaranteeing delivery is TCP's job. This is exactly why TCP and IP always work together.

HTTP — HyperText Transfer Protocol

What it does HTTP is the protocol that loads web pages. When you type a website address and press enter, HTTP is what fetches that page from the server and brings it to your browser.

Port HTTP runs on Port 80. The secure version HTTPS runs on Port 443.

How it works Your browser sends an HTTP request to the server asking for a page. The server finds that page and sends it back as an HTTP response. Your browser then displays it.

Important concept — Stateless HTTP does not remember anything between requests. Every time you click a link or refresh a page, the server treats it as a brand new request with no memory of what happened before. Websites use cookies to work around this and remember your login.

Real life example You ask a waiter for a menu. He brings it. You ask again a minute later and he has no memory of you — he brings it again fresh. That is stateless behavior.

FTP — File Transfer Protocol

What it does FTP is used to transfer files between your computer and a server. Web developers use it to upload their website files onto a hosting server.

Ports FTP uses two ports. Port 21 handles commands like login and navigation. Port 20 handles the actual file data being transferred.

Two modes In Active Mode the server connects back to the client to send data, which can get blocked by firewalls. In Passive Mode the client makes

both connections itself, which works better and is more commonly used today.

Real life example FTP is like a shared Google Drive folder but over a direct network connection. You connect, browse the folder, and upload or download files.

SMTP — Simple Mail Transfer Protocol

What it does SMTP sends your emails. The moment you click Send, SMTP carries your message from your mail app to the recipient's mail server.

Ports Port 25 is used for server to server communication. Port 587 is used when your email app submits a message to your mail server.

Most important point SMTP only sends email. It cannot receive or display emails. To receive emails, different protocols are used. POP3 downloads emails to your device and removes them from the server. IMAP keeps emails on the server and syncs them across all your devices.

Real life example SMTP is the postal truck that picks up your letter and delivers it to the post office. POP3 and IMAP are the ways you collect your own mail from your mailbox.

How TCP and IP Work Together

TCP and IP always work as a team. IP is responsible for finding the route and delivering packets across the network. TCP is responsible for making sure everything arrives correctly and completely. Without IP, data has no address or path. Without TCP, data can arrive missing pieces or out of order. Together they make internet communication both possible and reliable.

Difference Between HTTP, FTP, and SMTP

Protocol	Purpose
HTTP	Web browsing
FTP	File transfer
SMTP	Sending emails

CSMA/CD, CSMA/CA and Token Ring

CSMA/CD — Carrier Sense Multiple Access with Collision Detection

What it is CSMA/CD is a method used in wired Ethernet networks that controls how devices share a single cable. It solves the problem of two devices transmitting at the same time.

How it works Before sending, a device listens to the cable to check if it is free. If free, it starts transmitting. While sending, it keeps listening. If two devices transmit at the same time and their signals mix, a collision is detected. Both devices immediately stop, send a jam signal to alert everyone, and then each waits a random amount of time before trying again. This random wait is called the backoff period and it prevents the same devices from colliding again immediately.

Key points Collision is detected after it happens. The jam signal alerts all devices to stop. Random backoff prevents repeated collisions.

Real life example Two people accidentally speak at the same time in a meeting. Both stop, one says sorry loudly, and then each waits a random few seconds before trying again.

CSMA/CA — Carrier Sense Multiple Access with Collision Avoidance

What it is CSMA/CA is used in wireless Wi-Fi networks. In wireless, you cannot detect a collision while it is happening because your own signal drowns out everything else. So instead of detecting collisions, this method avoids them before they happen.

How it works Before sending, the device listens to check if the channel is free. When the channel becomes free, the device waits an additional random backoff time before transmitting. This extra wait stops multiple devices from jumping in at the same moment. For extra safety, a device can send a Request to Send signal to the access point, which replies with Clear to Send, reserving the channel before actual data is sent. After data is received, the receiver sends back an ACK to confirm delivery. No ACK means something went wrong and data is resent.

Key points Collision is avoided before it happens. Random backoff after channel clears is the main avoidance step. RTS and CTS reserve the channel. ACK confirms successful delivery.

Real life example In an online video call you wait for silence, then wait a couple more seconds before speaking. You might raise your hand and wait for the host to say go ahead before speaking. That is RTS and CTS in action.

CSMA/CD vs CSMA/CA — The Core Difference

CSMA/CD detects a collision after it happens and recovers from it. CSMA/CA prevents a collision before it happens. CD is used on wired networks where detection is possible. CA is used on wireless networks where detection is not reliable so avoidance is the only option.

Token Ring

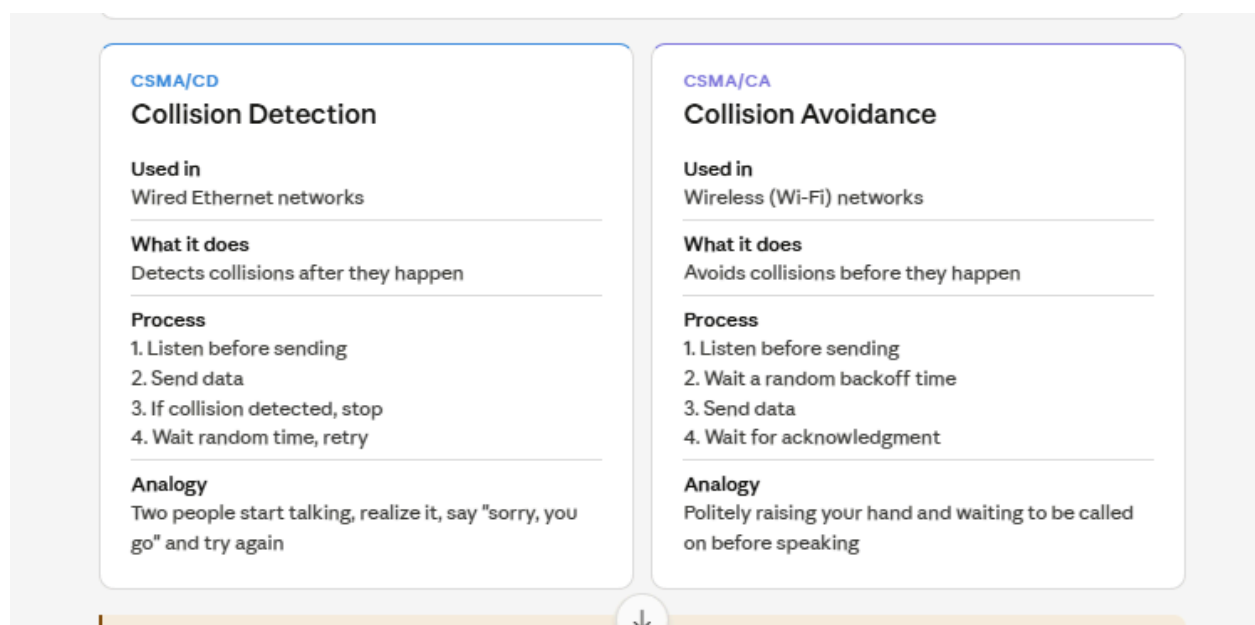
What it is Token Ring is a network where all devices are connected in a ring shape. A small signal called a token continuously travels around this ring. Only the device holding the token is allowed to send data at that moment.

How it works When a device wants to send data it waits for the token to reach it, grabs it, attaches its data, and sends it around the ring. Every device reads the destination address on the packet and only the intended

receiver copies the data. Once the packet travels all the way around and returns to the sender, the sender releases the token for the next device.

Key points Only one device transmits at a time so collisions are completely impossible. Every device gets a guaranteed fair turn. If the token is lost or any device fails, the entire network stops.

Real life example A talking stick passed around a meeting. Only the person holding the stick can speak. No two people ever talk at the same time. Token Ring works exactly the same way.



MAC Address, DHCP, DNS, WWW and Telnet

MAC Address — Media Access Control Address

What it is A MAC address is a permanent address that is physically built into every network card by the manufacturer. It is used to identify your device on a local network like your home Wi-Fi.

Format It is 48 bits long and written as 6 pairs like this — 00:1A:2B:3C:4D:5E. The first three pairs tell you who made the device and the last three pairs identify the device itself.

Key points It never changes because it is burned into the hardware. It works inside a local network. IP addresses work across different networks.

Real life example Your national ID number never changes no matter where you live — that is your MAC address. Your home address can change when you move — that is your IP address.

DHCP — Dynamic Host Configuration Protocol

What it is DHCP automatically gives an IP address to your device when you connect to a network. Without it someone would have to manually set an IP address on every single device which is not practical.

How it works DHCP follows four steps known as DORA.

Your device sends a Discover message asking if there is a DHCP server nearby. The server replies with an Offer giving you an available IP address. Your device sends a Request saying yes I want that address. The server sends an Acknowledge confirming the address is yours for a set period of time called the lease time.

Key points Lease time is how long you keep that IP address. When it expires the address goes back to the pool for someone else. This whole process happens automatically every time you connect to a network.

Real life example You check into a hotel and the receptionist gives you a room number for your stay. When you leave the room is given to the next guest. DHCP does the same thing with IP addresses.

DNS — Domain Name System

What it is DNS converts website names into IP addresses. Computers only understand numbers but humans remember names. DNS is the translator between the two.

How it works You type google.com in your browser. Your device checks its memory to see if it already knows the IP address. If not it asks your internet provider's DNS server. That server contacts other DNS servers one by one until it finds the correct IP address for google.com and sends it back. Your browser then uses that IP address to open the website.

Key points DNS uses Port 53. The whole translation happens in milliseconds. Your device saves recently visited addresses in a local cache to make future visits faster.

Real life example DNS is the phone book of the internet. You look up a name and it gives you the number. Without it you would have to memorize every website's IP address.

WWW — World Wide Web

What it is The World Wide Web is a collection of web pages that are linked together and accessed through a browser. It is very important to understand that the WWW and the internet are not the same thing. The internet is the physical network of cables and connections. The WWW is just one service that runs on top of that network.

How it works Every web page has its own address called a URL. When you enter a URL your browser uses HTTP to request that page from the server. The server sends back the page written in HTML and your browser displays it as the formatted page you see on screen.

Key points WWW was invented by Tim Berners-Lee in 1989. It is built on three things — HTML for page content, HTTP for transferring pages, and URLs for page addresses. Services like email and FTP also use the internet but they are not part of the [WWW](http://www.ietf.org/rfc/rfc2427.txt).

Real life example The internet is the road network. The WWW is the cars using those roads. Email and FTP are trucks and buses — they use the same roads but they are completely different services.

Telnet

What it is Telnet lets you remotely control another computer over a network using text commands. It is like sitting in front of that computer even though you are far away.

How it works You connect to a remote computer using its IP address on Port 23. Once connected you type commands and the remote computer runs them and shows you the results on your screen.

The big problem Telnet sends everything as plain text with zero encryption. Your username, password and every command you type can be seen by anyone monitoring the network. This makes it completely unsafe to use.

Replaced by SSH SSH does the exact same job as Telnet but encrypts everything so no one can read your data. This is why Telnet is no longer used in the real world.

Key points Telnet uses Port 23. No encryption at all. SSH on Port 22 is the safe modern replacement.

Real life example Telnet is like shouting your ATM PIN in a crowded room — it works but everyone can hear it. SSH is like sending it through a locked sealed envelope that only the receiver can open.

Key Points

IPv4 vs IPv6

- IPv4 = 32 bit address. IPv6 = 128 bit address.
- IPv4 written in decimal with dots. IPv6 written in hexadecimal with colons.
- IPv4 has 4.3 billion addresses. IPv6 has practically unlimited addresses.
- IPv6 created because IPv4 addresses were running out.
- IPv4 needs NAT. IPv6 does not need NAT.
- IPv6 has IPSec security built in. IPv4 has it optional.
- IPv4 header has 12 fields. IPv6 header has 8 fields — simpler and faster.
- Exam tip — Why was IPv6 invented? Always answer — address exhaustion.

TCP — Transmission Control Protocol

- TCP is connection oriented sets up connection before sending data.
- Uses Three Way Handshake SYN → SYN-ACK → ACK.
- Breaks data into segments and numbers them for correct order.

- If any segment is lost TCP automatically resends it.
- Reliable but slightly slower due to constant checking.
- Used where accuracy matters — web browsing, email, file transfer.
- Exam tip — Know the three way handshake steps in exact order.

IP — Internet Protocol

- IP handles addressing and routing of data.
- IP is connectionless — no setup before sending.
- Does not guarantee delivery, order or error correction.
- Just forwards packets toward destination using best available path.
- Different packets of same message can take different routes.
- IP handles routing. TCP handles reliability. They always work together.
- Exam tip — IP alone is unreliable. TCP makes the combination reliable.

HTTP — HyperText Transfer Protocol

- HTTP loads web pages through request and response model.
 - Port 80 for HTTP. Port 443 for HTTPS.
 - Browser sends request. Server sends page back as response.
 - HTTP is stateless server remembers nothing between requests.
 - Cookies are used to work around stateless behavior.
 - Exam tip Port 80 and stateless are the two most tested HTTP facts.
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FTP — File Transfer Protocol

- FTP transfers files between a client and a server.
- Port 21 for control commands. Port 20 for actual data transfer.
- Active Mode server initiates data connection back to client.
- Passive Mode client initiates both connections. Better behind firewalls.
- Passive mode is more commonly used today.
- FTP is the only protocol using two ports for two different jobs.

SMTP — Simple Mail Transfer Protocol

- SMTP is used only for sending emails.
- Port 25 for server to server. Port 587 for client to server.
- SMTP cannot receive emails sending only.
- POP3 receives emails and downloads them removing from server.
- IMAP receives emails and keeps them on server synced across devices.
- Exam tip — SMTP sends. POP3 and IMAP receive. Never mix these up.

MAC Address

- MAC address is a permanent 48 bit hardware address.
- Written as 6 pairs of hexadecimal digits — 00:1A:2B:3C:4D:5E.
- First 3 pairs identify the manufacturer.
- Last 3 pairs identify the specific device.
- Never changes — burned permanently into hardware.
- MAC address works within local network. IP address works across networks.
- Exam tip — MAC is permanent and physical. IP is logical and changeable.

CSMA/CD

- Used in wired Ethernet networks.
- Device listens before sending — if cable is busy it waits.
- If two devices transmit at same time a collision occurs.
- Both devices detect collision and stop immediately.
- Jam signal is sent to alert all devices on the network.
- Each device waits a random backoff time before retrying.
- Exam tip — CD means collision Detection. Detects after it happens. Wired only.

CSMA/CA

- Used in wireless Wi-Fi networks.
- Cannot detect collision in wireless so avoids it before it happens.
- Device listens and waits for channel to be free.
- Then waits additional random backoff time before transmitting.
- RTS sent to access point. Access point replies with CTS to reserve channel.
- Receiver sends ACK after getting data. No ACK means retransmit.
- Exam tip — CA means Collision Avoidance. Prevents before it happens. Wireless only.

CSMA/CD vs CSMA/CA

- CD detects collision after it happens. CA avoids collision before it happens.
- CD is used on wired networks. CA is used on wireless networks.
- CD reacts to collisions. CA prevents collisions.
- Exam tip — This comparison is the single most tested point from these two topics.

Token Ring

- All devices connected in a ring shape.
- A token continuously travels around the ring.
- Only the device holding the token can transmit data.
- Device waits for token, grabs it, attaches data and sends it around ring.
- Only the intended receiver copies the data. Others pass it along.
- Sender releases token after data returns successfully.
- Zero collisions possible because only one device transmits at a time.
- If token is lost or any device fails the entire network stops.
- Exam tip — No collisions is the biggest advantage. Single point of failure is the biggest weakness.

DHCP — Dynamic Host Configuration Protocol

- DHCP automatically assigns IP addresses when a device joins a network.
- Follows four steps called DORA.
- D — Discover. Device asks if any DHCP server is available.
- O — Offer. Server offers an available IP address.
- R — Request. Device accepts the offered IP address.
- A — Acknowledge. Server confirms and sets the lease time.
- Lease time is how long the device keeps that IP address.
- When lease expires IP returns to pool for another device.
- Exam tip — Memorize DORA steps in exact order. Always comes in exams.

DNS — Domain Name System

- DNS translates domain names into IP addresses.
- Runs on Port 53.
- Device checks local cache first before contacting DNS server.
- If not cached it contacts ISP resolver → root server → TLD server → authoritative server.
- Final server provides the correct IP address.
- DNS cache stores recent addresses to speed up future requests.
- Exam tip — DNS is the phone book of the internet. Port 53. Name to IP translation.

WWW — World Wide Web

- WWW is a system of interlinked web pages accessed through a browser.
- WWW is not the same as the internet.
- Internet is the physical infrastructure. WWW is one service running on it.
- Built on three things — HTML for content, HTTP for transfer, URLs for addresses.
- Invented by Tim Berners-Lee in 1989.
- Email and FTP also use the internet but are not part of [WWW](http://www.).
- Exam tip — Internet and WWW are not the same. This distinction is commonly tested.

Telnet

- Telnet allows remote access and control of another computer over a network.
- Runs on Port 23.
- Sends everything in plain text — zero encryption.
- Username, password and all commands are visible to anyone on the network.
- Completely insecure for modern use.
- Replaced by SSH which does the same job but with full encryption.
- SSH runs on Port 22.
- Exam tip — Telnet equals no encryption equals insecure. SSH is the secure replacement. Port 23 vs Port 22.